

Digital Forensics

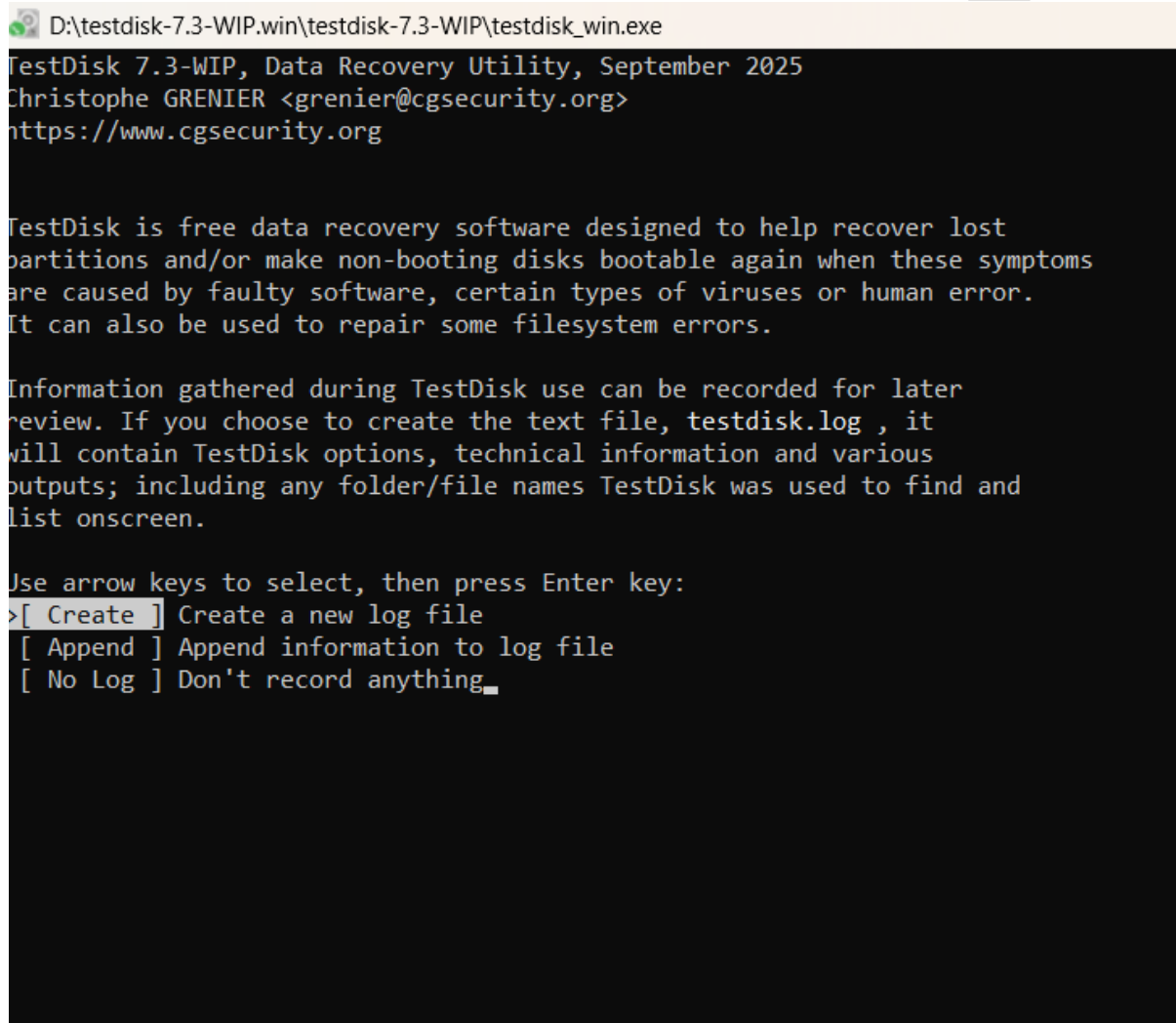
Ex.No:2 Recover deleted or damaged files from a storage device using Test Disk

Date:

Description:

TestDisk step by step to recover a missing partition and repair a corrupted one.

Log creation

A screenshot of a Windows command prompt window titled "D:\testdisk-7.3-WIP.win\testdisk-7.3-WIP\testdisk_win.exe". The window shows the TestDisk 7.3-WIP interface. It displays the version "TestDisk 7.3-WIP, Data Recovery Utility, September 2025" and the author "Christophe GRENIER <grenier@cgsecurity.org>" with the website "https://www.cgsecurity.org". A descriptive paragraph follows: "TestDisk is free data recovery software designed to help recover lost partitions and/or make non-booting disks bootable again when these symptoms are caused by faulty software, certain types of viruses or human error. It can also be used to repair some filesystem errors." Another paragraph explains logging: "Information gathered during TestDisk use can be recorded for later review. If you choose to create the text file, testdisk.log, it will contain TestDisk options, technical information and various outputs; including any folder/file names TestDisk was used to find and list onscreen." The prompt then asks the user to "Use arrow keys to select, then press Enter key:" and shows three options: "> [Create] Create a new log file", "[Append] Append information to log file", and "[No Log] Don't record anything_". The "Create" option is currently selected.

```
D:\testdisk-7.3-WIP.win\testdisk-7.3-WIP\testdisk_win.exe
TestDisk 7.3-WIP, Data Recovery Utility, September 2025
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

TestDisk is free data recovery software designed to help recover lost
partitions and/or make non-booting disks bootable again when these symptoms
are caused by faulty software, certain types of viruses or human error.
It can also be used to repair some filesystem errors.

Information gathered during TestDisk use can be recorded for later
review. If you choose to create the text file, testdisk.log, it
will contain TestDisk options, technical information and various
outputs; including any folder/file names TestDisk was used to find and
list onscreen.

Use arrow keys to select, then press Enter key:
> [ Create ] Create a new log file
  [ Append ] Append information to log file
  [ No Log ] Don't record anything_
```

- Choose Create to instruct Testdisk to create a [log file](#) containing technical information and messages, unless you have a reason to append data to the [log](#) or you execute TestDisk from read only media and must create the [log](#) elsewhere.
- Choose None if you do not want messages and details of the process to be written into a [log file](#) (useful if for example Testdisk was started from a read-only location).
- Press Enter to proceed.

Disk selection

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All hard drives should be detected and listed with the correct size by TestDisk:

```
D:\testdisk-7.3-WIP.win\testdisk-7.3-WIP\testdisk_win.exe
TestDisk 7.3-WIP, Data Recovery Utility, September 2025
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  TestDisk is free software, and
comes with ABSOLUTELY NO WARRANTY.

Select a media and choose 'Proceed' using arrow keys:
>Disk \\.PhysicalDrive0 - 512 GB / 476 GiB - MTFDKBA512TFK-1BC1AABHA
Drive F: - 408 MB / 389 MiB - Msft Virtual DVD-ROM

>[Proceed ] [ Quit ]

Note: Serial number 0000_0000_0000_0001_00A0_7522_3C07_EC77.
Disk capacity must be correctly detected for a successful recovery.
If a disk listed above has an incorrect size, check HD jumper settings and BIOS
detection, and install the latest OS patches and disk drivers.
```

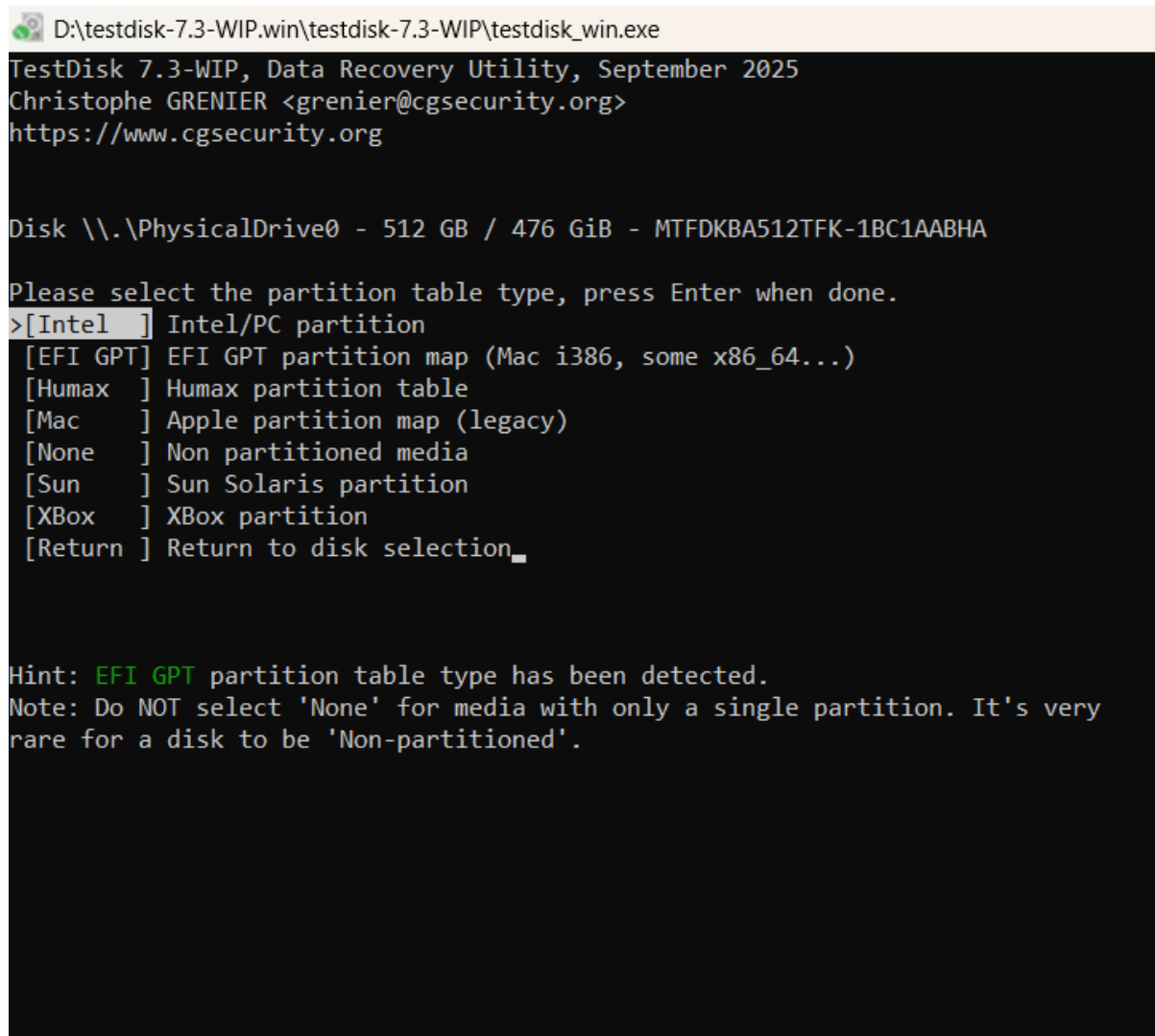
- Use up/down arrow keys to select your hard drive with the lost partition/s.
- Press Enter to Proceed.

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X If available, use raw device `/dev/rdisk*` instead of `/dev/disk*` for faster data transfer.

Partition table type selection

TestDisk displays the partition table types.



```
D:\testdisk-7.3-WIP.win\testdisk-7.3-WIP\testdisk_win.exe
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Disk \\.\\PhysicalDrive0 - 512 GB / 476 GiB - MTFDKBA512TFK-1BC1AABHA

Please select the partition table type, press Enter when done.
>[Intel ] Intel/PC partition
[EFI GPT] EFI GPT partition map (Mac i386, some x86_64...)
[Humax  ] Humax partition table
[Mac    ] Apple partition map (legacy)
[None   ] Non partitioned media
[Sun    ] Sun Solaris partition
[XBox   ] XBox partition
[Return] Return to disk selection_

Hint: EFI GPT partition table type has been detected.
Note: Do NOT select 'None' for media with only a single partition. It's very
rare for a disk to be 'Non-partitioned'.
```

- Select the partition table type - usually the default value is the correct one as TestDisk auto-detects the partition table type.
- Press Enter to Proceed.

Current partition table status

TestDisk displays the menus (also see [TestDisk Menu Items](#)).

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```
D:\testdisk-7.3-WIP.win\testdisk-7.3-WIP\testdisk_win.exe
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https://www.cgsecurity.org

Disk \\.\PhysicalDrive0 - 512 GB / 476 GiB - MTFDKBA512TFK-1BC1AABHA
CHS 62260 255 63 - sector size=512

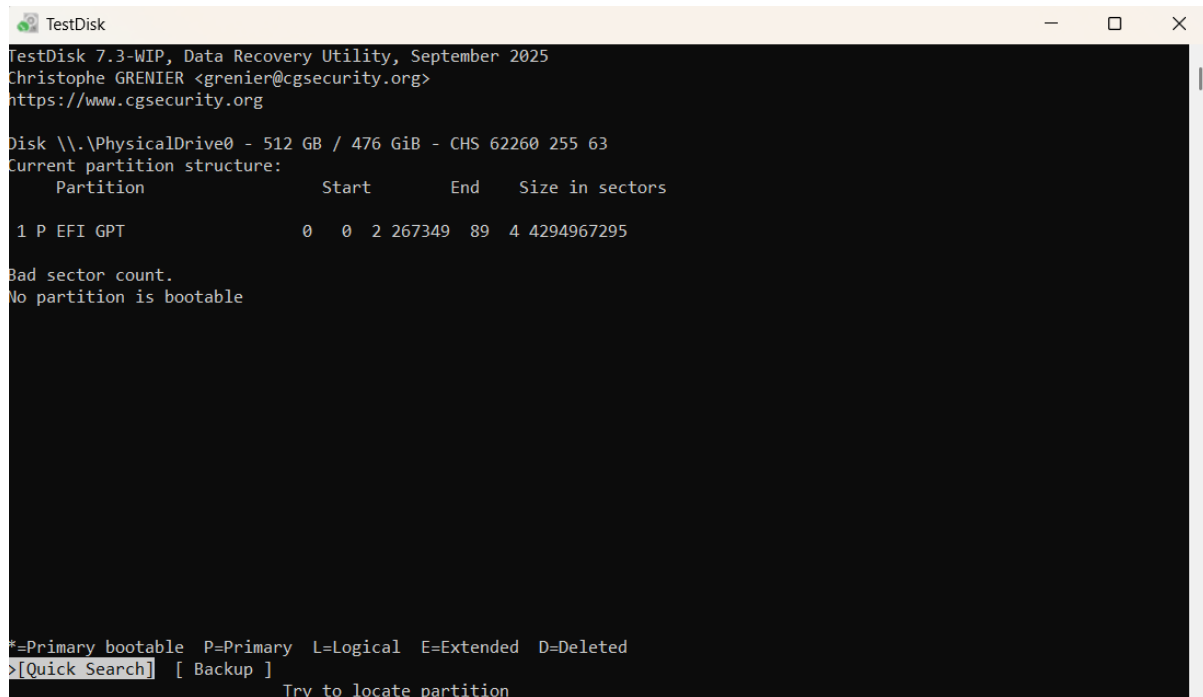
>[ Analyse ] Analyse current partition structure and search for lost partitions
[ Advanced ] Filesystem Utils
[ Geometry ] Change disk geometry
[ Options ] Modify options
[ Quit ] Return to disk selection

Note: Correct disk geometry is required for a successful recovery. 'Analyse'
process may give some warnings if it thinks the logical geometry is mismatched.
```

- Use the default menu "Analyse" to check your current partition structure and search for lost partitions.
- Confirm at Analyse with Enter to proceed.

Now, your current partition structure is listed. Examine your current partition structure for missing partitions and errors.

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```
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Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

Disk \\.\PhysicalDrive0 - 512 GB / 476 GiB - CHS 62260 255 63
Current partition structure:

Partition              Start          End          Size in sectors

1 P EFI GPT            0 0 2 267349  89 4 4294967295

Bad sector count.
No partition is bootable

*=Primary bootable  P=Primary  L=Logical  E=Extended  D=Deleted
>[Quick Search]  [ Backup ]
Try to locate partition
```

The first partition is listed twice which points to a corrupted partition or an invalid partition table entry.

Invalid NTFS boot points to a faulty NTFS boot sector, so it's a corrupted filesystem.

Only one logical partition (label Partition 2) is available in the extended partition. One logical partition is missing.

- Confirm at **Quick Search** to proceed.

Quick Search for partitions

TestDisk displays the first results in real time.



(click on thumb to display the image).

During the **Quick Search**, TestDisk has found two partitions including the missing logical partition labeled **Partition 3**.

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```
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Disk \\.\PhysicalDrive0 - 512 GB / 476 GiB - CHS 62260 255 63
Partition      Start      End      Size in sectors
s* FAT32        0 32 33    33 69 36    532480 [EFI System Partition] [SYSTEM]
P HPFS - NTFS   35953 100 52 36077 13 32    1986560
D HPFS - NTFS   62185 131 27 62259 181 2    1191936 [Windows RE tools]
D HPFS - NTFS   62185 131 27 62260 121 5    1204224 [Windows RE tools]

Structure: Ok. Use Up/Down Arrow keys to select partition.
Use Left/Right Arrow keys to CHANGE partition characteristics:
#=Primary bootable P=Primary L=Logical E=Extended D=Deleted
Keys A: add partition, L: load backup, T: change type, P: list files,
Enter: to continue
FAT32, blocksize=4096, 272 MB / 260 MiB
```

- Highlight this partition and press **p** to list your files (to go back to the previous display, press q to Quit, Files listed in red are deleted entries).

All directories and data are correctly listed.

- Press Enter to proceed.

Save the partition table or search for more partitions?

```
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Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org
* FAT32        0 32 33    33 69 36    532480 [EFI System Partition] [SYSTEM]
Directory /EFI
>drwxr-xr-x    0    0    0 20-Dec-2022 10:31 .
drwxr-xr-x    0    0    0 20-Dec-2022 10:31 ..
drwxr-xr-x    0    0    0 12-Jan-2026 16:38 Microsoft
drwxr-xr-x    0    0    0 20-Dec-2022 10:31 Boot
drwxr-xr-x    0    0    0 8-Jul-2024 12:13 HP

Next
Use Left arrow to go back, Right to change directory, 'h' to hide deleted files
'q' to quit, ':' to select the current file, 'a' to select all files
'C' to copy the selected files, 'c' to copy the current file
```

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- **When all partitions are available** and data correctly listed, you should go to the menu **Write** to save the partition structure. The menu Extd Part gives you the opportunity to decide if the extended partition will use all available disk space or only the required (minimal) space.
- **Since a partition, the first one, is still missing**, highlight the menu **Deeper Search** (if not done automatically already) and press Enter to proceed.

A partition is still missing: Deeper Search

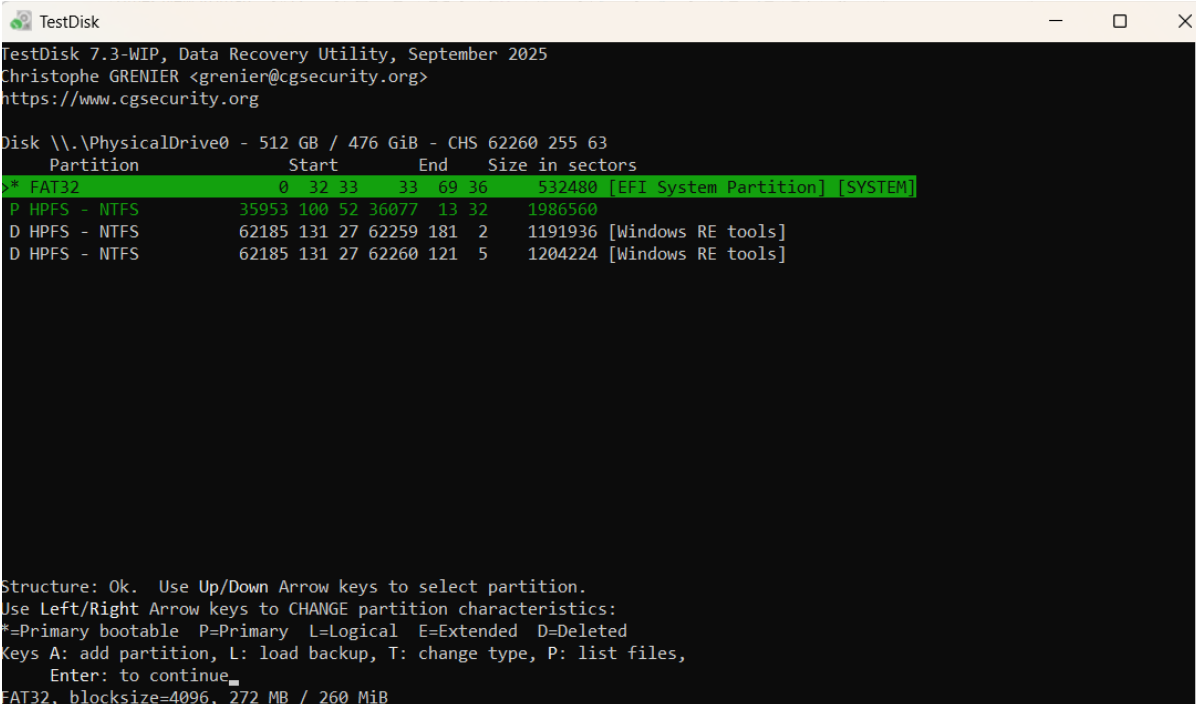
Deeper Search will also search for FAT32 backup boot sector, NTFS backup boot superblock, ext2/ext3 backup superblock to detect more partitions,

it will scan each cylinder  (click on thumb).

After the Deeper Search, the results are displayed as follows:

The first partition "**Partition 1**" was found by using backup boot sector. In the last line of your display, you can read the message "**NTFS found using backup sector!**" and the size of your partition. The "partition 2" is displayed twice with different size.

Partitions listed as D(leted) will not be recovered if you let them listed as deleted. Both partitions are listed with status **D** for deleted, because they overlap each other. You need to identify which partition to recover.



```
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Christophe GRENIER <grenier@cgsecurity.org>
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Disk \\.\PhysicalDrive0 - 512 GB / 476 GiB - CHS 62260 255 63
Partition      Start      End      Size in sectors
* FAT32         0 32 33    33 69 36    532480 [EFI System Partition] [SYSTEM]
P HPFS - NTFS   35953 100 52 36077 13 32    1986560
D HPFS - NTFS   62185 131 27 62259 181 2    1191936 [Windows RE tools]
D HPFS - NTFS   62185 131 27 62260 121 5    1204224 [Windows RE tools]

Structure: Ok. Use Up/Down Arrow keys to select partition.
Use Left/Right Arrow keys to CHANGE partition characteristics:
*=Primary bootable P=Primary L=Logical E=Extended D=Deleted
Keys A: add partition, L: load backup, T: change type, P: list files,
Enter: to continue
FAT32, blocksize=4096, 272 MB / 260 MiB
```

- Highlight the first partition Partition 2 and press **p** to list its data.

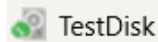
The file system of the upper logical partition (label Partition 2) is  (click on thumb).

- Press q for Quit to go back to the previous display.
- Let this partition Partition 2 with a damaged file system marked as D(deleted).
- Highlight the second partition Partition 2 below

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- Press p to list its files. qq

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```
TestDisk 7.3-WIP, Data Recovery Utility, September 2025

Please select a destination where /EFI/ and any files below will be copied.
Keys: Arrow keys to select another directory
      C when the destination is correct
      Q to quit
Directory D:\testdisk-7.3-WIP.win\testdisk-7.3-WIP
>drwxrwx--- 197609 197609      0 26-Aug-2026 20:06 .
drwxrwx--- 197609 197609      0 26-Aug-2026 20:05 ..
TestDisk 7.3-WIP, Data Recovery Utility, September 2025

Please select a destination where /EFI/ and any files below will be copied.
Keys: Arrow keys to select another directory
      C when the destination is correct
      Q to quit
Directory D:\testdisk-7.3-WIP.win\testdisk-7.3-WIP
>drwxrwx--- 197609 197609      0 26-Aug-2026 20:06 .
drwxrwx--- 197609 197609      0 26-Aug-2026 20:05 ..
drwxrwx--- 197609 197609      0 26-Aug-2026 20:05 63
drwxrwx--- 197609 197609      0 26-Aug-2026 20:05 platforms
-rwxrwx--- 197609 197609    220 26-Aug-2026 20:05 AUTHORS.txt
-rwxrwx--- 197609 197609   18326 26-Aug-2026 20:05 COPYING.txt
-rwxrwx--- 197609 197609    120 26-Aug-2026 20:05 INFO
-rwxrwx--- 197609 197609   20989 26-Aug-2026 20:05 NEWS.txt
-rwxrwx--- 197609 197609  6762631 26-Aug-2026 20:05 Qt5Core.dll
-rwxrwx--- 197609 197609  7646324 26-Aug-2026 20:05 Qt5Gui.dll
-rwxrwx--- 197609 197609  7935180 26-Aug-2026 20:05 Qt5Widgets.dll
-rwxrwx--- 197609 197609    350 26-Aug-2026 20:05 THANKS.txt
-rwxrwx--- 197609 197609     40 26-Aug-2026 20:05 VERSION.txt
-rwxrwx--- 197609 197609  4210072 26-Aug-2026 20:05 cygwf-2.dll
-rwxrwx--- 197609 197609   823067 26-Aug-2026 20:05 cyggcc_s-1.dll
-rwxrwx--- 197609 197609  1315500 26-Aug-2026 20:05 cygiconv-2.dll
-rwxrwx--- 197609 197609  1964540 26-Aug-2026 20:05 cygjpeg-8.dll
-rwxrwx--- 197609 197609  3462774 26-Aug-2026 20:05 cygncursesw-10.dll
-rwxrwx--- 197609 197609  3274927 26-Aug-2026 20:05 cygwin1.dll
-rwxrwx--- 197609 197609   289410 26-Aug-2026 20:05 cygz.dll
-rwxrwx--- 197609 197609    504 26-Aug-2026 20:05 documentation.html
-rwxrwx--- 197609 197609   368752 26-Aug-2026 20:05 fidetify_win.exe
-rwxrwx--- 197609 197609    37788 26-Aug-2026 20:05 iconv.dll
-rwxrwx--- 197609 197609   116814 26-Aug-2026 20:05 libbz2-1.dll
-rwxrwx--- 197609 197609  6788902 26-Aug-2026 20:05 libewf-2.dll
-rwxrwx--- 197609 197609   869036 26-Aug-2026 20:05 libfreetype-6.dll
-rwxrwx--- 197609 197609  1072581 26-Aug-2026 20:05 libgcc_s_dw2-1.dll
-rwxrwx--- 197609 197609  1674649 26-Aug-2026 20:05 libglib-2.0-0.dll
-rwxrwx--- 197609 197609  7038385 26-Aug-2026 20:05 libharfbuzz-0.dll
-rwxrwx--- 197609 197609   239106 26-Aug-2026 20:05 libintl-8.dll
-rwxrwx--- 197609 197609   776885 26-Aug-2026 20:05 libjpeg-62.dll
-rwxrwx--- 197609 197609   620258 26-Aug-2026 20:05 libpcre2-16-0.dll
-rwxrwx--- 197609 197609   675308 26-Aug-2026 20:05 libpcre2-8-0.dll
```

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It works, your files are listed, you have found the correct partition!

- Use the left/right arrow to navigate into your folders and watch your files for more verification

Note: FAT directory listing is limited to 10 clusters - some files may not appear but it doesn't affect recovery.

- Press q for Quit to go back to the previous display.
- The available status are Primary, * bootable, Logical and Deleted.

Using the left/right arrow keys, change the status of the selected partition from D(eleted) to **L(ogical)**. This way you will be able to recover this partition.

Hint: read [How to recognize primary and logical partitions?](#)

Note: If a partition is listed *(bootable) but if you don't boot from this partition, you can change it to **P**Primary partition.

- Press Enter to proceed.

Partition table recovery

It's now possible to write the new partition structure.

Note: The extended partition is automatically set. TestDisk recognizes this using the different partition structure.

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```
TestDisk
TestDisk 7.3-WIP, Data Recovery Utility, September 2025

Please select a destination where /EFI/ and any files below will be copied.
Keys: Arrow keys to select another directory
      C when the destination is correct
      Q to quit
Directory D:\testdisk-7.3-WIP.win\testdisk-7.3-WIP
>drwxrwx--- 197609 197609      0 26-Aug-2026 20:06 .
drwxrwx--- 197609 197609      0 26-Aug-2026 20:05 ..
TestDisk 7.3-WIP, Data Recovery Utility, September 2025
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org
* FAT32      0 32 33      33 69 36      532480 [EFI System Partition] [SYSTEM]
Directory /EFI
>drwxr-xr-x 0 0 0 20-Dec-2022 10:31 .
drwxr-xr-x 0 0 0 20-Dec-2022 10:31 ..
drwxr-xr-x 0 0 0 12-Jan-2026 16:38 Microsoft
drwxr-xr-x 0 0 0 20-Dec-2022 10:31 Boot
drwxr-xr-x 0 0 0 8-Jul-2024 12:13 HP
```

- If **all partitions are listed** and only in this case, confirm at **Write** with Enter, y and OK.

Now, the partitions are registered in the partition table.

NTFS Boot sector recovery

The boot sector of the first partition named Partition 1 is still damaged. It's time to fix it. The status of the NTFS boot sector is bad and the backup boot sector is valid. Boot sectors are not identical.

```
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https://www.cgsecurity.org

Disk \\.\PhysicalDrive0 - 512 GB / 476 GiB - CHS 62260 255 63
Partition      Start      End      Size in sectors
2 P HPFS - NTFS 62185 131 27 62260 121 5 1204224 [Windows RE tools]

Boot sector
Status: OK

Backup boot sector
Status: OK

Sectors are not identical.

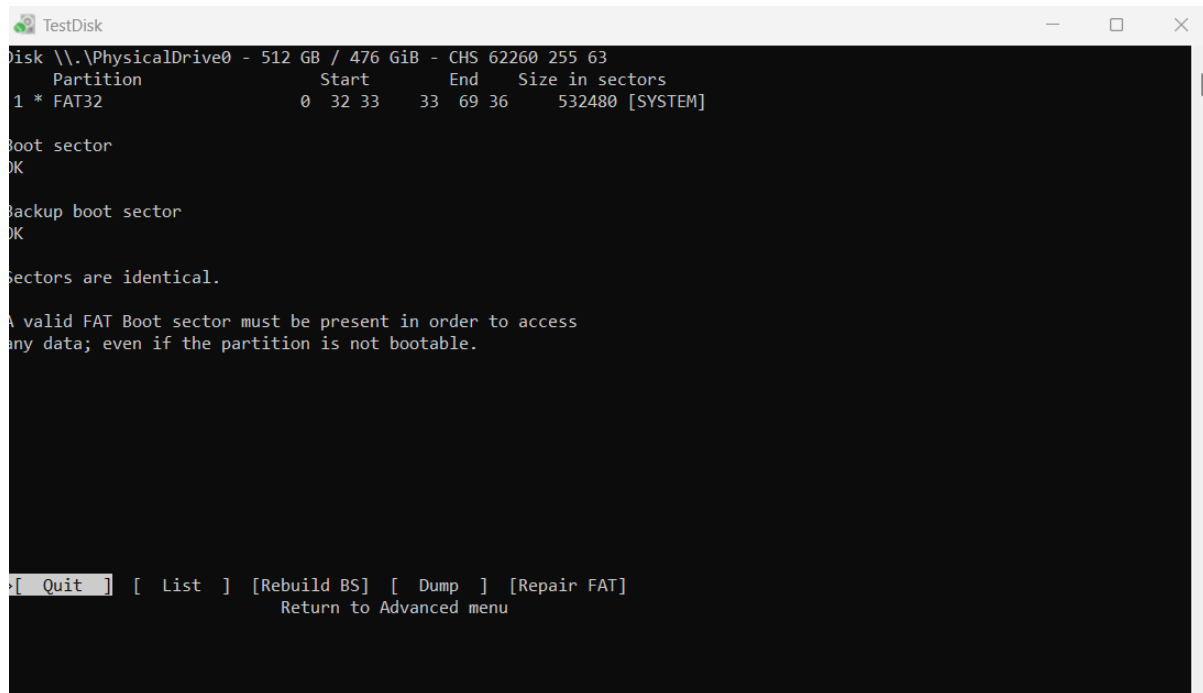
A valid NTFS Boot sector must be present in order to access
any data; even if the partition is not bootable.

>[ Quit ] [ List ] [Org. BS] [Backup BS] [Rebuild BS] [ Dump ]
Return to Advanced menu
```

- To copy the backup of the boot sector over the boot sector, select **Backup BS**, validate with Enter, use y to confirm and next OK.

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More information about repairing your boot sector under [TestDisk Menu Items](#). The following message is displayed:



```
TestDisk
Disk \\.\PhysicalDrive0 - 512 GB / 476 GiB - CHS 62260 255 63
Partition      Start      End      Size in sectors
1 * FAT32       0 32 33   33 69 36   532480 [SYSTEM]

Boot sector
OK

Backup boot sector
OK

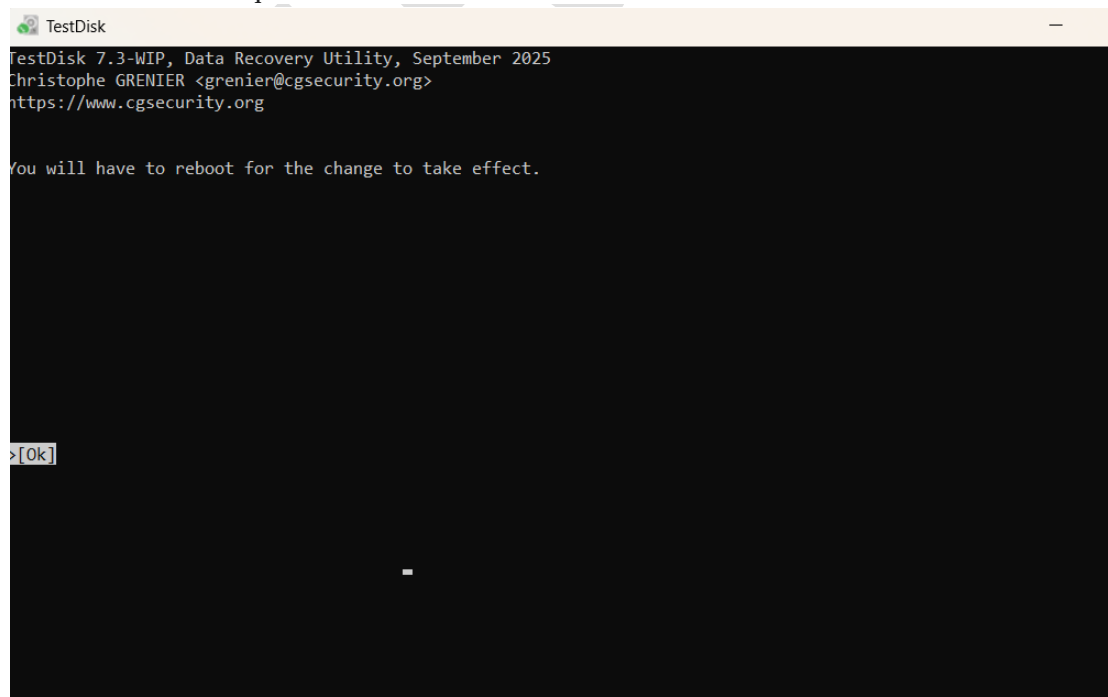
Sectors are identical.

A valid FAT Boot sector must be present in order to access
any data; even if the partition is not bootable.

[ Quit ] [ List ] [Rebuild BS] [ Dump ] [Repair FAT]
Return to Advanced menu
```

The boot sector and its backup are now both OK and identical: the NTFS boot sector has been successfully recovered.

- Press Enter to quit.



```
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You will have to reboot for the change to take effect.

>[Ok]
```

- TestDisk displays ***You have to restart your Computer to access your data*** so press `Enter` a last time and reboot your computer.